Query Optimization

MCQ

1. The main goal of query optimization in a DBMS is to:
A. Minimize the number of tuples returned
B. Choose the cheapest query execution plan
C. Maximize query execution time
D. Reduce data redundancy

2- Which operation should be applied as early as possible in a query execution plan to reduce intermediate results?

A. Join
B. Projection
C. Selection
D. Aggregation

3. Which operation is generally the most expensive in query processing?

A. Selection
B. Projection
C. Join
D. Rename

4. In query optimization, pushing projection early mainly helps to:

A. Reduce number of tuples
B. Reduce number of attributes
C. Eliminate joins
D. Increase selectivity

5. Which join method is most efficient when an index exists on the join attribute?

A. Nested loop join
B. Hash join
C. Indexed nested loop join
D. Sort-merge join

6. Which of the following is a benefit of join reordering?

A. Eliminates the need for indexes
B. Reduces intermediate result sizes

C. Removes duplicate tuples
D. Simplifies SQL syntax

7. In relational algebra optimization, projections can be pushed down the tree provided that:

A. They include all attributes
B. They do not remove attributes needed later
C. They follow a join
D. They are applied after selection

8. Which statistic is MOST useful for estimating the cost of a selection operation?

A. Number of tables
B. Attribute cardinality
C. SQL query length
D. Join order

9. Which of the following best describes selectivity?

A. Ratio of disk blocks to tuples
B. Fraction of tuples satisfying a condition
C. Number of joins in a query
D. Cost of projection

Problem Solving:

Given the following tables:
Teams (teamid, tname, currentcoach)
Players (playerid, pname, teamid, points)
Coaches (coachid, cname)

Table	Pages	Column Info	Indexes
Teams	100		
Players	500	Rating has values [1, 10]	Clustered on `rating` (height = 1, 100 leaf pages) Clustered on `playerid` (height = 1, 50 leaf pages)
Coaches	200		

Answer the following questions about the following query:

```
SELECT pname, cname
FROM Teams t
INNER JOIN Players p ON t.teamid = p.teamid
INNER JOIN Coaches c ON t.currentcoach = c.coachid
WHERE p.rating > 5
ORDER BY p.playerid;
```

1) What is the selectivity of the p.rating> 5 condition?

2) What **single table access** plans will advance to the next pass?

- a- Full scan on Teams
- b- Full scan on Players
- c- Index scan on Players.rating
- d- Index scan on Players.playerid
- e- Full scan on Coaches

3) Which of the following 2 table plans will advance from the second pass to the third pass? Assume that the I/Os provided is optimal for those tables joined in that order and also assume that the output of every plan is not sorted.

- a- Teams Join Players (1000 I/Os)
- b- Players Join Teams (1500 I/Os)
- c- Players Join Coaches (2000 I/Os)
- d- Coaches Join Players (800 I/Os)
- e- Teams Join Coaches (500 I/Os)
- f- Coaches Join Teams (750 I/Os)

4) Which of the following 3 table plans will be considered during the third pass? Again assume that the I/Os provided is optimal for joining the tables in that order and that none of the outputs are sorted.

- a- (Teams Join Players) Join Coaches - 10,000 I/Os
- b- Coaches Join (Teams Join Players) - 8,000 I/Os
- c- (Players Join Teams) Join Coaches - 9,000 I/Os
- d- (Players Join Coaches) Join Teams - 12,000 I/Os
- e- (Teams Join Coaches) Join Players - 15,000, I/Os

5) Which 3 table plan will be used?

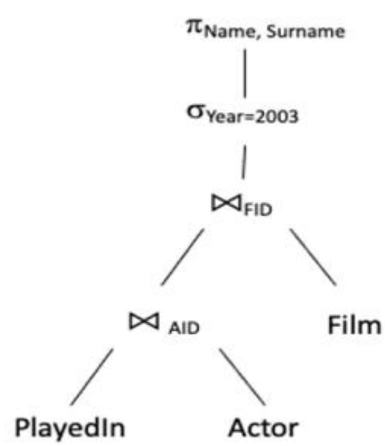
Given the following Tables:

Film (FID, Title, Year)
Actor (AID, Name, Surname)
PlayedIn (FID, AID, Character)

Given the following Query:

```
SELECT A.Name, A.Surname
FROM Actor A
JOIN PlayedIn P ON A.AID = P.AID
JOIN Film F ON P.FID = F.FID
WHERE F.Year = 2003;
```

The above query is expressed by the following Tree**: Create** an **optimized** version of the search tree which minimizes access to rows and columns.



Concurrency Control and Recovery

MCQ:

1- Which property of a transaction ensures that concurrent execution produces the same result as serial execution?

A. Atomicity
B. Consistency
C. Isolation
D. Durability

2- Which type of conflict occurs when two transactions access the same data item and at least one access is a write?

A. Read–Read
B. Read–Write
C. Write–Write
D. Both B and C

3- Which problem occurs when a transaction reads uncommitted data from another transaction?

A. Lost update
B. Dirty read
C. Phantom read
D. Incorrect summary

4- A deadlock occurs when:

A. Two transactions read the same data
B. Transactions wait indefinitely for each other's locks
C. A transaction fails before commit
D. Locks are released early

5- The main purpose of recovery techniques is to:

A. Speed up transaction execution
B. Prevent deadlocks
C. Restore the database to a consistent state after failure
D. Reduce concurrency

6- Write-Ahead Logging (WAL) requires that:

A. Data pages are written before log pages
B. Log records are written before data pages
C. Logs are written after commit
D. All of the above

7- Which transactions are redone during recovery?

A. All transactions
B. Uncommitted transactions
C. Committed transactions after last checkpoint
D. Transactions that caused deadlock

8- Which log record indicates successful completion of a transaction?

A. START
B. ABORT
C. COMMIT
D. CHECKPOINT

9- Which property of ACID is mainly ensured by recovery mechanisms?

A. Atomicity and Durability
B. Consistency and Isolation
C. Isolation and Durability
D. Atomicity and Isolation

Problem Solving

1- Given The following schedule that consists of two transactions, T1 and T2; determine whether the schedule is conflict serializable or not. **Justify Your Answer.** **Hint**: Use the Dependency Graph

T1:	R(A), W(A)	
T2:		R(A), W(A), R(B), W(B)

2- Given The following schedule that consists of two transactions, T1 and T2; determine whether the schedule is conflict serializable or not. **Justify Your Answer.** **<u>Hint</u>:** Use the Dependency Graph

T1:	R(A), W(A)		R(B)
T2:		R(A), W(A), R(B), W(B)	

3- Is the following Schedule possible under the 2PL (2-Phase Locking)? S() Means acquiring a shared lock, X() means acquiring an exclusive lock, U() means releasing a lock. **Justify Your Answer.**

T1:	X(A)	X(C)		U(A)		U(C)		
T2:			S(B)				U(B)	
T3:					S(A)			U(A)

4- Is the above schedule possible under the strict 2PL (Strict 2-Phase Locking)? **Justify Your Answer.**